

Brutal Practice Paper B44 - Electric Current, Perimeter & Area, Physical & Chemical Changes, Rational Numbers

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. Battery A is made of 4 cells in series, each giving 1.5 V. Battery B is made of 3 cells in series, each giving 2 V. By how many volts do the two batteries differ? [\[Electric Current & its Effects\]](#)
 - (A) 2 V
 - (B) 0 V (they are equal)
 - (C) 6 V
 - (D) 3 V
 - (E) 12 V
 - (F) 0.5 V

2. A rectangular field measures 84 m by 36 m. A jogger runs all the way around its boundary 3 times. What total distance does the jogger cover, in kilometres? [\[Perimeter & Area\]](#)
 - (A) 1.2 km
 - (B) 7.2 km
 - (C) 0.24 km
 - (D) 2.16 km
 - (E) 0.36 km
 - (F) 0.72 km
 - (G) 720 km

3. Rusting needs BOTH air (oxygen) and moisture. Five iron nails are kept in: dry air, boiled water with an oil layer on top, plain tap water, moist air, and oil only. In how many will rust form? [\[Physical & Chemical Changes\]](#)
 - (A) 4 nails
 - (B) only the dry-air nail
 - (C) 0 nails
 - (D) 1 nail
 - (E) 2 nails
 - (F) 3 nails

4. Add the rational numbers $-\frac{3}{4}$ and $\frac{5}{6}$ and give the answer in its lowest terms. [\[Rational Numbers\]](#)
 - (A) $\frac{1}{12}$
 - (B) $-\frac{7}{12}$
 - (C) $\frac{2}{10}$
 - (D) $-\frac{1}{12}$
 - (E) $\frac{8}{12}$
 - (F) $\frac{7}{12}$
 - (G) $\frac{1}{2}$

5. A heater coil gives out 240 J of heat in 8 seconds at a steady rate. How much heat does it give in one full minute? [\[Electric Current & its Effects\]](#)
- (A) 240 J
 - (B) 1800 J
 - (C) 3600 J
 - (D) 900 J
 - (E) 1440 J
 - (F) 30 J
6. A square park has a side of 45 m. A path 2 m wide runs along the inside of its boundary. What is the area of the path? [\[Perimeter & Area\]](#)
- (A) 172 square metres
 - (B) 2025 square metres
 - (C) 1681 square metres
 - (D) 184 square metres
 - (E) 344 square metres
 - (F) 88 square metres
 - (G) 360 square metres
7. A blacksmith heats iron until it glows, hammers it into a horseshoe, then lets it cool. The shaping into a horseshoe is which kind of change, and can it be undone? [\[Physical & Chemical Changes\]](#)
- (A) Both physical and chemical
 - (B) Chemical change, reversible
 - (C) Physical change, and it is reversible
 - (D) Chemical change, irreversible
 - (E) Chemical, because heat was used
 - (F) Neither physical nor chemical
8. Work out $7/9 - (-2/3)$ and give the answer in lowest terms. [\[Rational Numbers\]](#)
- (A) $9/13$
 - (B) $13/9$
 - (C) $1/3$
 - (D) $-13/9$
 - (E) $1/9$
 - (F) $5/9$
9. An electromagnet with 50 turns of wire can just lift 200 g of iron. If lifting power is proportional to the number of turns, how many turns are needed to lift 700 g? [\[Electric Current & its Effects\]](#)
- (A) 350 turns
 - (B) 175 turns
 - (C) 25 turns
 - (D) 250 turns
 - (E) 100 turns
 - (F) 140 turns
 - (G) 700 turns

- 10.** A circular garden has a radius of 21 m. Find the length of fencing needed to go once around it. (Use $\pi = 22/7$.) [\[Perimeter & Area\]](#)
- (A) 462 m
 - (B) 42 m
 - (C) 264 m
 - (D) 154 m
 - (E) 1386 m
 - (F) 66 m
 - (G) 132 m
- 11.** From 250 g of an impure copper sulphate solution, crystallisation produces 180 g of pure crystals. What percentage of the starting mass is recovered as pure crystals? [\[Physical & Chemical Changes\]](#)
- (A) 30%
 - (B) 36%
 - (C) 28%
 - (D) 18%
 - (E) 70%
 - (F) 180%
 - (G) 72%
- 12.** Multiply $-8/15$ by $-5/4$ and give the answer in lowest terms. [\[Rational Numbers\]](#)
- (A) $-3/2$
 - (B) $40/60$
 - (C) $1/3$
 - (D) $4/3$
 - (E) $-2/3$
 - (F) $2/3$
- 13.** Three appliances drawing 3 A, 4 A and 2 A run together on one circuit. A fuse must be rated just above the total current. Of the fuses 5 A, 10 A and 13 A, which is the smallest safe one? [\[Electric Current & its Effects\]](#)
- (A) 13 A
 - (B) 10 A
 - (C) 9 A
 - (D) 5 A
 - (E) 15 A
 - (F) 20 A
 - (G) 8 A
- 14.** A circular garden has a radius of 21 m. Find the area of grass inside it. (Use $\pi = 22/7$.) [\[Perimeter & Area\]](#)
- (A) 693 square metres
 - (B) 132 square metres
 - (C) 462 square metres
 - (D) 138 square metres
 - (E) 1386 square metres
 - (F) 441 square metres
 - (G) 2772 square metres

15. Galvanisation coats iron with a layer of zinc to stop rusting. If coating needs 4 g of zinc per square metre, how much zinc is needed to coat 15 square metres of sheet? [\[Physical & Chemical Changes\]](#)

- (A) 19 g
- (B) 60 g
- (C) 11 g
- (D) 64 g
- (E) 45 g
- (F) 600 g
- (G) 30 g

16. Divide $-6/7$ by $3/14$ and give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) -4
- (B) $1/4$
- (C) $-9/98$
- (D) 12
- (E) -2
- (F) 4

17. An electric iron uses 1000 J of electrical energy every second. How many kilojoules does it use in 5 minutes? [\[Electric Current & its Effects\]](#)

- (A) 5 kJ
- (B) 100 kJ
- (C) 1500 kJ
- (D) 60 kJ
- (E) 3000 kJ
- (F) 300 kJ

18. A triangular plot has a base of 40 m and a height of 25 m. Laying turf costs Rs 12 per square metre. What is the total cost of turfing it? [\[Perimeter & Area\]](#)

- (A) Rs 600
- (B) Rs 1000
- (C) Rs 500
- (D) Rs 5000
- (E) Rs 12000
- (F) Rs 3000
- (G) Rs 6000

19. Which one of these is a CHEMICAL change: melting wax, dissolving sugar in water, burning a wax candle, boiling water, breaking a glass, or folding paper? [\[Physical & Chemical Changes\]](#)

- (A) Evaporating water
- (B) Burning a wax candle
- (C) Breaking a glass
- (D) Folding paper
- (E) Boiling water
- (F) Dissolving sugar in water

20. Write $-27/45$ in standard form (lowest terms with a positive denominator). [\[Rational Numbers\]](#)

- (A) $5/-3$
- (B) $-27/45$
- (C) $-3/5$
- (D) $3/5$
- (E) $-9/15$
- (F) $-5/3$
- (G) $-3/-5$

21. A device needs exactly 9 V to run. Using cells of 1.5 V each, how many cells are required and how must they be joined? [\[Electric Current & its Effects\]](#)

- (A) 9 cells in series
- (B) 6 cells, the positive of each joined to the negative of the next
- (C) 13 cells in series
- (D) 6 cells, all positives joined together
- (E) 4 cells in series
- (F) 6 cells joined side by side (parallel)
- (G) 3 cells, positives joined to positives

22. A parallelogram has a base of 18 cm and a height of 7 cm. A triangle has the same base and the same height. By how much is the parallelogram's area larger than the triangle's? [\[Perimeter & Area\]](#)

- (A) 126 square cm
- (B) 189 square cm
- (C) 63 square cm
- (D) 31.5 square cm
- (E) 25 square cm
- (F) 252 square cm
- (G) 90 square cm

23. Rust is iron joined with oxygen, so a rusted nail is heavier than before. If a 20 g nail gains 15% in mass after fully rusting, what is its new mass? [\[Physical & Chemical Changes\]](#)

- (A) 20 g
- (B) 35 g
- (C) 23 g
- (D) 30 g
- (E) 3 g
- (F) 17 g
- (G) 26 g

24. Which is greater, $-5/8$ or $-7/12$? [\[Rational Numbers\]](#)

- (A) both equal 0
- (B) $-8/5$ is greater
- (C) $-7/12$ is greater
- (D) they are equal
- (E) $-12/7$ is greater
- (F) $-1/2$ is greater

- 25.** In an electric bell, the hammer strikes the gong 5 times every second while the switch is pressed. How many times does it strike in 2.5 minutes of ringing? [\[Electric Current & its Effects\]](#)
- (A) 150 strikes
 - (B) 300 strikes
 - (C) 750 strikes
 - (D) 1500 strikes
 - (E) 12 strikes
 - (F) 75 strikes
- 26.** An L-shaped room is made by cutting a 3 m by 4 m corner out of a 10 m by 8 m rectangle. What is the floor area of the room? [\[Perimeter & Area\]](#)
- (A) 70 square metres
 - (B) 80 square metres
 - (C) 12 square metres
 - (D) 60 square metres
 - (E) 56 square metres
 - (F) 68 square metres
- 27.** Burning magnesium ribbon forms white magnesium oxide ash. If 6 g of magnesium gives 10 g of white ash, how many grams of oxygen combined with the magnesium? [\[Physical & Chemical Changes\]](#)
- (A) 4 g
 - (B) 16 g
 - (C) 6 g
 - (D) 2 g
 - (E) 8 g
 - (F) 10 g
 - (G) 60 g
- 28.** What rational number lies exactly halfway between $\frac{1}{4}$ and $\frac{1}{2}$? [\[Rational Numbers\]](#)
- (A) $\frac{2}{6}$
 - (B) $\frac{1}{3}$
 - (C) $\frac{3}{4}$
 - (D) $\frac{3}{8}$
 - (E) $\frac{1}{8}$
 - (F) $\frac{3}{6}$
- 29.** A filament bulb uses 60 J of energy each second to give a certain brightness. An LED giving the same brightness uses only 9 J each second. What percentage of energy does the LED save? [\[Electric Current & its Effects\]](#)
- (A) 85%
 - (B) 60%
 - (C) 69%
 - (D) 15%
 - (E) 9%
 - (F) 6.6%
 - (G) 51%

30. A wire of length 88 cm is bent to form a circle. What is the radius of the circle? (Use $\pi = 22/7$.) [\[Perimeter & Area\]](#)

- (A) 22 cm
- (B) 44 cm
- (C) 11 cm
- (D) 14 cm
- (E) 28 cm
- (F) 4 cm
- (G) 7 cm

31. Photosynthesis is a chemical change in which a leaf makes glucose. If a leaf makes 0.8 g of glucose every hour in sunlight, how much glucose does it make in 6 hours of daylight? [\[Physical & Chemical Changes\]](#)

- (A) 3.6 g
- (B) 6.8 g
- (C) 48 g
- (D) 5.6 g
- (E) 1.4 g
- (F) 4.8 g
- (G) 0.13 g

32. What is the reciprocal of $-4/9$? [\[Rational Numbers\]](#)

- (A) 1
- (B) $-4/9$
- (C) $9/4$
- (D) $-9/-4$
- (E) -1
- (F) $-9/4$
- (G) $4/9$

33. Three identical 1.5 V cells are joined in a series chain, but one cell is put in the wrong way round (reversed). What is the net voltage of the chain? [\[Electric Current & its Effects\]](#)

- (A) 1.5 V
- (B) 6 V
- (C) 4.5 V
- (D) 2 V
- (E) 3 V
- (F) 0.5 V

34. The same 88 cm wire is now bent into a square instead of a circle. What is the area of that square? [\[Perimeter & Area\]](#)

- (A) 22 square cm
- (B) 256 square cm
- (C) 121 square cm
- (D) 196 square cm
- (E) 441 square cm
- (F) 484 square cm
- (G) 88 square cm

- 35.** Which of these changes can be reversed: rusting of iron, burning of paper, melting of ice, cooking of an egg, curdling of milk, or ripening of fruit? [\[Physical & Chemical Changes\]](#)
- (A) Cooking of an egg
 - (B) Burning of paper
 - (C) Burning of wood
 - (D) Melting of ice
 - (E) Curdling of milk
 - (F) Rusting of iron
 - (G) Ripening of fruit
- 36.** Which number must be added to $-\frac{11}{13}$ to give a result of zero (its additive inverse)? [\[Rational Numbers\]](#)
- (A) 0
 - (B) $\frac{11}{13}$
 - (C) 1
 - (D) $\frac{13}{11}$
 - (E) $-\frac{13}{11}$
 - (F) $-\frac{11}{13}$
 - (G) $\frac{11}{-13}$
- 37.** One heater coil releases 18 J of heat per second. A second coil releases heat at 1.5 times that rate. How much heat does the second coil give in 40 seconds? [\[Electric Current & its Effects\]](#)
- (A) 18 J
 - (B) 27 J
 - (C) 108 J
 - (D) 1080 J
 - (E) 540 J
 - (F) 1620 J
- 38.** A rectangle and a square have the same perimeter, 64 cm. The rectangle is 20 cm long. By how much does the square's area exceed the rectangle's area? [\[Perimeter & Area\]](#)
- (A) 240 square cm
 - (B) 40 square cm
 - (C) 16 square cm
 - (D) 32 square cm
 - (E) 96 square cm
 - (F) 4 square cm
 - (G) 256 square cm
- 39.** When baking soda is added to vinegar, bubbles of gas fizz out. This is called a chemical change mainly because: [\[Physical & Chemical Changes\]](#)
- (A) the change can be reversed easily
 - (B) a new substance (the gas) is formed
 - (C) the liquids are simply mixed
 - (D) only the shape changes
 - (E) no new substance is formed
 - (F) it is just dissolving

40. Among $-\frac{1}{2}$, $\frac{3}{4}$, $-\frac{3}{4}$ and $\frac{1}{4}$, which number is the smallest? [\[Rational Numbers\]](#)

- (A) $-\frac{3}{4}$
- (B) 0
- (C) $-\frac{1}{4}$
- (D) $\frac{3}{4}$
- (E) $\frac{1}{4}$
- (F) $-\frac{1}{2}$
- (G) $\frac{1}{2}$

41. Four bulbs are connected in a single series loop. If one bulb's filament breaks, how many bulbs stay lit, and why? [\[Electric Current & its Effects\]](#)

- (A) 3 bulbs, only the broken one goes off
- (B) all 4 glow brighter
- (C) 0 bulbs stay lit, because a break in a series loop stops all current
- (D) 4 bulbs, none are affected
- (E) 1 bulb stays lit
- (F) 3 bulbs share the current
- (G) 2 bulbs stay lit

42. A floor measuring 6 m by 4.5 m is to be covered with square tiles of side 30 cm. How many tiles are needed? [\[Perimeter & Area\]](#)

- (A) 30 tiles
- (B) 600 tiles
- (C) 200 tiles
- (D) 300 tiles
- (E) 270 tiles
- (F) 90 tiles

43. In slow crystallisation, a solution forms 5 g of crystals every 10 minutes at a steady rate. How long will it take to crystallise 35 g? [\[Physical & Chemical Changes\]](#)

- (A) 7 minutes
- (B) 50 minutes
- (C) 350 minutes
- (D) 175 minutes
- (E) 70 minutes
- (F) 14 minutes
- (G) 35 minutes

44. Evaluate $\frac{2}{3} + (-\frac{4}{5}) + \frac{1}{15}$ and give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $-\frac{1}{5}$
- (B) $\frac{1}{15}$
- (C) $-\frac{11}{15}$
- (D) $-\frac{1}{15}$
- (E) $-\frac{1}{3}$
- (F) $\frac{11}{15}$

45. A compass needle near a current-carrying wire turns by 20 degrees when the current is 2 A. If the turn is proportional to the current, what current would turn it by 50 degrees (small-angle case)? [\[Electric Current & its Effects\]](#)

- (A) 25 A
- (B) 4 A
- (C) 10 A
- (D) 70 A
- (E) 7 A
- (F) 5 A
- (G) 2.5 A

46. A path 1 m wide runs all around the OUTSIDE of a rectangular pool 12 m by 5 m. Find the area of the path. [\[Perimeter & Area\]](#)

- (A) 98 square metres
- (B) 48 square metres
- (C) 76 square metres
- (D) 38 square metres
- (E) 34 square metres
- (F) 17 square metres

47. Painting an iron gate keeps out air and moisture and so prevents rusting. A gate of area 8 square metres needs 250 mL of paint per square metre for each coat, and gets two coats. How much paint is needed in total? [\[Physical & Chemical Changes\]](#)

- (A) 0.5 L
- (B) 4 L
- (C) 8 L
- (D) 16 L
- (E) 2000 L
- (F) 40 L

48. A number multiplied by $-\frac{5}{6}$ gives $\frac{10}{3}$. What is the number? [\[Rational Numbers\]](#)

- (A) -16
- (B) $-\frac{25}{9}$
- (C) 4
- (D) 2
- (E) $\frac{1}{4}$
- (F) -4

49. A bulb uses 100 J of energy every second. If it is switched on for 6 hours, how much energy does it use, in kilojoules? [\[Electric Current & its Effects\]](#)

- (A) 2160 kJ
- (B) 21600 kJ
- (C) 216 kJ
- (D) 600 kJ
- (E) 36 kJ
- (F) 1080 kJ

- 50.** A flat ring lies between two circles sharing the same centre: outer radius 14 cm and inner radius 7 cm. Find the area of the ring. (Use $\pi = 22/7$.) [\[Perimeter & Area\]](#)
- (A) 462 square cm
 - (B) 21 square cm
 - (C) 308 square cm
 - (D) 231 square cm
 - (E) 154 square cm
 - (F) 616 square cm
- 51.** Souring of milk into curd is a chemical change. If 1 L of milk sets into 0.9 kg of curd, how much curd forms from 5 L of milk? [\[Physical & Chemical Changes\]](#)
- (A) 0.9 kg
 - (B) 5 kg
 - (C) 5.9 kg
 - (D) 4 kg
 - (E) 9 kg
 - (F) 4.5 kg
- 52.** Simplify $(-2/3) \times (9/14) \div (3/7)$ and give the answer in lowest terms. [\[Rational Numbers\]](#)
- (A) $3/7$
 - (B) $-9/14$
 - (C) $-7/3$
 - (D) $-3/7$
 - (E) -1
 - (F) $-1/3$
- 53.** An electromagnet's strength depends on turns multiplied by current. Coil A has 100 turns at 2 A; Coil B has 50 turns at 5 A. Which is stronger, and by how much (in ampere-turns)? [\[Electric Current & its Effects\]](#)
- (A) Coil A, by 50 ampere-turns
 - (B) Coil B, by 250 ampere-turns
 - (C) Coil A, by 100 ampere-turns
 - (D) Coil A, by 250 ampere-turns
 - (E) they are equal
 - (F) Coil B, by 150 ampere-turns
 - (G) Coil B, by 50 ampere-turns
- 54.** A rectangular garden 50 m by 30 m is to be fenced with 4 rounds of wire. The wire costs Rs 8 per metre. Find the total cost of the wire. [\[Perimeter & Area\]](#)
- (A) Rs 1500
 - (B) Rs 640
 - (C) Rs 5120
 - (D) Rs 6400
 - (E) Rs 1280
 - (F) Rs 2560

55. A chemical change often shows signs such as heat, light, sound, gas, or a colour change. When a firecracker bursts, how many of these signs appear: heat, light, sound, gas? [\[Physical & Chemical Changes\]](#)

- (A) 1
- (B) 3
- (C) only sound and gas
- (D) 0
- (E) 2
- (F) 4 (all four signs)
- (G) only heat and light

56. Which statement is true about $-6/8$, $6/-8$ and $-9/12$? [\[Rational Numbers\]](#)

- (A) none of these equal $-3/4$
- (B) only $6/-8$ equals $-3/4$
- (C) only $-6/8$ equals $-3/4$
- (D) only $-9/12$ equals $-3/4$
- (E) $-6/8$ and $6/-8$ only
- (F) $-9/12$ and $-6/8$ only
- (G) all of these are equal to $-3/4$

57. A fuse wire melts when the current through it goes above 5 A. Two appliances of 2 A each are already on; a third appliance of 2 A is switched on. Does the fuse melt? [\[Electric Current & its Effects\]](#)

- (A) Yes, the total is exactly 5 A
- (B) Yes, the total 6 A is above 5 A, so it melts
- (C) No, each appliance is only 2 A
- (D) No, the total 4 A is below 5 A
- (E) No, the fuse never melts
- (F) Yes, but only because of one appliance
- (G) No, the total is 2 A

58. A rectangle is 16 cm by 9 cm. A triangle is drawn using one full 16 cm side as its base and its tip touching the opposite side. What fraction of the rectangle's area is the triangle? [\[Perimeter & Area\]](#)

- (A) $1/8$
- (B) $1/2$
- (C) $1/4$
- (D) $2/3$
- (E) $3/4$
- (F) $1/3$
- (G) 1

59. Stainless steel resists rust because it contains chromium. If 100 kg of a stainless steel contains 18 kg of chromium and 8 kg of nickel (and the rest is iron), how much iron does it contain? [\[Physical & Chemical Changes\]](#)

- (A) 90 kg
- (B) 82 kg
- (C) 26 kg
- (D) 64 kg
- (E) 18 kg
- (F) 74 kg
- (G) 92 kg

60. Subtract the sum of $-\frac{2}{5}$ and $\frac{3}{10}$ from 1. What do you get? [\[Rational Numbers\]](#)

- (A) $\frac{11}{10}$
- (B) $\frac{10}{11}$
- (C) $-\frac{1}{10}$
- (D) $\frac{1}{10}$
- (E) $\frac{21}{10}$
- (F) $-\frac{9}{10}$
- (G) $\frac{9}{10}$

61. A heater raises the temperature of water by 5 degrees Celsius each minute. How long does it take to raise the water from 20 degrees Celsius to 80 degrees Celsius? [\[Electric Current & its Effects\]](#)

- (A) 5 minutes
- (B) 60 minutes
- (C) 10 minutes
- (D) 15 minutes
- (E) 12 minutes
- (F) 16 minutes
- (G) 80 minutes

62. The area of a square tile is 196 square cm. What is its perimeter? [\[Perimeter & Area\]](#)

- (A) 784 cm
- (B) 49 cm
- (C) 56 cm
- (D) 28 cm
- (E) 14 cm
- (F) 196 cm
- (G) 112 cm

63. Melting and evaporation are physical changes. If 200 g of ice melts to water and then 50 g of that water evaporates, how much liquid water remains? [\[Physical & Chemical Changes\]](#)

- (A) 50 g
- (B) 175 g
- (C) 250 g
- (D) 200 g
- (E) 150 g
- (F) 0 g
- (G) 100 g

64. How many rational numbers lie between $\frac{2}{7}$ and $\frac{5}{7}$? [\[Rational Numbers\]](#)

- (A) infinitely many
- (B) only $\frac{3}{7}$ and $\frac{4}{7}$
- (C) 3
- (D) a finite number
- (E) none
- (F) 2

- 65.** A battery stores 5400 J of energy. A bulb draws 3 J of energy every second. For how many minutes will the battery keep the bulb lit? [\[Electric Current & its Effects\]](#)
- (A) 1800 minutes
 - (B) 30 minutes
 - (C) 90 minutes
 - (D) 300 minutes
 - (E) 27 minutes
 - (F) 18 minutes
 - (G) 15 minutes
- 66.** A room measures 5 m by 4 m. A carpet is laid leaving a 0.5 m border of bare floor all around. Find the area of the carpet. [\[Perimeter & Area\]](#)
- (A) 6 square metres
 - (B) 10 square metres
 - (C) 16 square metres
 - (D) 8 square metres
 - (E) 9 square metres
 - (F) 20 square metres
 - (G) 12 square metres
- 67.** The setting of cement is a chemical change that cannot be undone. One bag of cement (50 kg) plasters 4 square metres of wall. How many whole bags are needed to plaster 26 square metres? [\[Physical & Chemical Changes\]](#)
- (A) 7 bags
 - (B) 8 bags
 - (C) 5 bags
 - (D) 104 bags
 - (E) 6 bags
 - (F) 4 bags
- 68.** Solve for x: $x/5 = -12/15$. [\[Rational Numbers\]](#)
- (A) $-4/5$
 - (B) 4
 - (C) -36
 - (D) -2
 - (E) 12
 - (F) -4
 - (G) 36
- 69.** A toy needs 6 V to work. You have only 1.5 V cells and you join 3 of them in series. By how much are you short, and how many more cells do you need? [\[Electric Current & its Effects\]](#)
- (A) Exactly 6 V; no more cells
 - (B) Short by 4.5 V; 3 more cells
 - (C) Short by 1.5 V; 1 more cell is needed
 - (D) Short by 3 V; 2 more cells
 - (E) Short by 0.5 V; 1 more cell
 - (F) Short by 1.5 V; 2 more cells
 - (G) Over by 1.5 V; remove a cell

70. Two square fields have sides of 25 m and 15 m. By how much does the larger field's area exceed the smaller field's area? [\[Perimeter & Area\]](#)

- (A) 400 square metres
- (B) 100 square metres
- (C) 160 square metres
- (D) 200 square metres
- (E) 40 square metres
- (F) 625 square metres
- (G) 850 square metres

71. Digestion of food is a chemical change. If your body chemically breaks down 75% of a 320 g meal, how many grams are broken down? [\[Physical & Chemical Changes\]](#)

- (A) 80 g
- (B) 245 g
- (C) 320 g
- (D) 75 g
- (E) 200 g
- (F) 160 g
- (G) 240 g

72. The product of two rational numbers is 1. If one of them is $-\frac{7}{9}$, what is the other? [\[Rational Numbers\]](#)

- (A) $-\frac{7}{9}$
- (B) $\frac{7}{9}$
- (C) 1
- (D) $\frac{9}{7}$
- (E) $-\frac{9}{7}$
- (F) -1
- (G) $-\frac{16}{9}$

73. A heating coil is made of 60 turns of nichrome wire, with each turn 4 cm long. What total length of wire is used, in metres? [\[Electric Current & its Effects\]](#)

- (A) 2.4 m
- (B) 3.6 m
- (C) 15 m
- (D) 240 m
- (E) 0.24 m
- (F) 24 m
- (G) 64 m

74. A wheel has a radius of 35 cm. How far does it travel along the ground in one full turn? (Use $\pi = \frac{22}{7}$.) [\[Perimeter & Area\]](#)

- (A) 155 cm
- (B) 440 cm
- (C) 220 cm
- (D) 110 cm
- (E) 1.1 m
- (F) 70 cm
- (G) 3850 cm

75. When you heat sugar, it first melts (a physical change) and then turns black and smells burnt. The blackening is a chemical change because: [\[Physical & Chemical Changes\]](#)

- (A) a new substance (carbon) with new properties is formed
- (B) the sugar only changed shape
- (C) the colour change is not important
- (D) no new substance formed
- (E) it is a physical change
- (F) it can be cooled back to sugar

76. Which of these rational numbers lies closest to 0 on the number line: $-1/3$, $1/4$, $-1/5$, or $2/7$? [\[Rational Numbers\]](#)

- (A) $-1/3$
- (B) $1/4$
- (C) $-1/5$
- (D) they are all equal
- (E) $1/2$
- (F) $2/7$
- (G) 0

77. Each time an electric bell 'rings' it makes 8 hammer strikes. During a fire drill the bell rings 15 times. How many strikes are made in all? [\[Electric Current & its Effects\]](#)

- (A) 120 strikes
- (B) 23 strikes
- (C) 1815 strikes
- (D) 100 strikes
- (E) 240 strikes
- (F) 60 strikes

78. To cover a distance of 44 m, how many complete turns must a wheel of radius 35 cm make? (Use $\pi = 22/7$.) [\[Perimeter & Area\]](#)

- (A) 10 turns
- (B) 40 turns
- (C) 2 turns
- (D) 100 turns
- (E) 200 turns
- (F) 20 turns

79. An unprotected iron pipe loses 0.6 mm of thickness to rust each year. A 5 mm thick pipe is unsafe once MORE than half its thickness is gone. After how many years does that happen? [\[Physical & Chemical Changes\]](#)

- (A) 2 years
- (B) 8 years
- (C) 3 years
- (D) 6 years
- (E) 5 years
- (F) 4 years

80. Divide $-5/12$ by $-5/12$. What is the result? [\[Rational Numbers\]](#)

- (A) $-25/144$
- (B) 1
- (C) -1
- (D) $25/144$
- (E) $-5/12$
- (F) $5/12$
- (G) 0

81. In a house, 5 identical bulbs are each connected so they get full mains voltage (in parallel), and each draws 0.5 A. What total current is drawn from the mains? [\[Electric Current & its Effects\]](#)

- (A) 2 A
- (B) 5 A
- (C) 0.5 A
- (D) 25 A
- (E) 10 A
- (F) 0.1 A
- (G) 2.5 A

82. A parallelogram has an area of 96 square cm and a base of 12 cm. Find its height. [\[Perimeter & Area\]](#)

- (A) 16 cm
- (B) 8 cm
- (C) 12 cm
- (D) 1152 cm
- (E) 84 cm
- (F) 6 cm
- (G) 4 cm

83. Cloud formation, when water vapour turns into tiny droplets, is a physical change. If 360 g of vapour condenses and each gram forms 20 tiny droplets, how many droplets form? [\[Physical & Chemical Changes\]](#)

- (A) 720 droplets
- (B) 340 droplets
- (C) 7200 droplets
- (D) 72000 droplets
- (E) 3600 droplets
- (F) 380 droplets
- (G) 18 droplets

84. A rope of length $7/2$ m is cut into pieces, each $7/8$ m long. How many full pieces can be cut? [\[Rational Numbers\]](#)

- (A) 4 pieces
- (B) 8 pieces
- (C) 3 pieces
- (D) 28 pieces
- (E) 2 pieces
- (F) 16 pieces
- (G) 6 pieces

- 85.** Heater X gives 2000 J each second and runs for 30 seconds. Heater Y gives 1500 J each second and runs for 50 seconds. Which gives more heat, and by how much? [\[Electric Current & its Effects\]](#)
- (A) Heater X, by 500 J
 - (B) Heater Y, by 75000 J
 - (C) they are equal
 - (D) Heater Y, by 5000 J
 - (E) Heater X, by 60000 J
 - (F) Heater Y, by 15000 J
 - (G) Heater X, by 15000 J
- 86.** A hall is a rectangle 20 m by 12 m, with a semicircular stage of diameter 14 m added along one short side. Find the total floor area. (Use $\pi = 22/7$.) [\[Perimeter & Area\]](#)
- (A) 287 square metres
 - (B) 154 square metres
 - (C) 308 square metres
 - (D) 240 square metres
 - (E) 317 square metres
 - (F) 394 square metres
 - (G) 331 square metres
- 87.** Burning is a chemical change that releases energy. If burning 1 g of dry wood releases 16 kJ of energy, how much energy is released by burning 250 g of wood? [\[Physical & Chemical Changes\]](#)
- (A) 250 kJ
 - (B) 400 kJ
 - (C) 4000 kJ
 - (D) 40000 kJ
 - (E) 16 kJ
 - (F) 2000 kJ
- 88.** Evaluate $(-3/4) - (-5/8) - (1/2)$ and give the answer in lowest terms. [\[Rational Numbers\]](#)
- (A) $-1/8$
 - (B) $3/8$
 - (C) $5/8$
 - (D) $-7/8$
 - (E) $-5/4$
 - (F) $-3/8$
 - (G) $-5/8$
- 89.** A current of 6 A enters a junction and splits equally into 3 identical branches. What current flows in each branch? [\[Electric Current & its Effects\]](#)
- (A) 18 A
 - (B) 2 A
 - (C) 1 A
 - (D) 6 A
 - (E) 3 A
 - (F) 0.5 A
 - (G) 9 A

- 90.** A square lawn of side 28 m has a circular flower bed of radius 7 m in the middle. How much lawn area is left around the bed? (Use $\pi = 22/7$.) [\[Perimeter & Area\]](#)
- (A) 588 square metres
 - (B) 154 square metres
 - (C) 728 square metres
 - (D) 616 square metres
 - (E) 630 square metres
 - (F) 784 square metres
 - (G) 462 square metres
- 91.** Which PAIR contains BOTH physical changes: melting ice and burning coal; boiling water and dissolving salt; rusting iron and cooking rice; or digestion and curdling of milk? [\[Physical & Chemical Changes\]](#)
- (A) boiling water and dissolving salt
 - (B) photosynthesis and respiration
 - (C) rusting iron and cooking rice
 - (D) digestion and curdling of milk
 - (E) rusting and ripening
 - (F) burning paper and frying an egg
 - (G) melting ice and burning coal
- 92.** By what number should $-8/15$ be multiplied to get $2/5$? [\[Rational Numbers\]](#)
- (A) $-16/75$
 - (B) $-4/3$
 - (C) $-3/4$
 - (D) $3/4$
 - (E) $4/3$
 - (F) $-2/3$
 - (G) $-1/4$
- 93.** A bulb is rated to glow for about 1000 hours before its filament burns out. If it is used 4 hours every day, for how many days will it last? [\[Electric Current & its Effects\]](#)
- (A) 996 days
 - (B) 250 days
 - (C) 100 days
 - (D) 333 days
 - (E) 500 days
 - (F) 25 days
 - (G) 4000 days
- 94.** A rectangular sheet of card is 24 cm by 18 cm. The largest possible square is cut from it. What is the area of the leftover piece? [\[Perimeter & Area\]](#)
- (A) 324 square cm
 - (B) 432 square cm
 - (C) 90 square cm
 - (D) 72 square cm
 - (E) 144 square cm
 - (F) 108 square cm
 - (G) 216 square cm

- 95.** Common salt is obtained from seawater by evaporation and crystallisation. If 2000 L of seawater yields 70 kg of salt, how much salt comes from 5000 L of the same seawater? [\[Physical & Chemical Changes\]](#)
- (A) 87.5 kg
 - (B) 140 kg
 - (C) 175 kg
 - (D) 100 kg
 - (E) 70 kg
 - (F) 350 kg
- 96.** Which is the largest of these rational numbers: $-\frac{2}{3}$, $-\frac{3}{5}$, $-\frac{4}{7}$, $-\frac{5}{9}$? [\[Rational Numbers\]](#)
- (A) they are all equal
 - (B) $-\frac{5}{9}$
 - (C) $-\frac{2}{3}$
 - (D) $-\frac{9}{5}$
 - (E) $-\frac{4}{7}$
 - (F) $-\frac{1}{2}$
 - (G) $-\frac{3}{5}$
- 97.** A 4.5 V battery (three 1.5 V cells in series) drives a coil that releases 9 J of heat each second. How much heat does the coil release in 4 minutes? [\[Electric Current & its Effects\]](#)
- (A) 4320 J
 - (B) 216 J
 - (C) 1080 J
 - (D) 9 J
 - (E) 540 J
 - (F) 36 J
 - (G) 2160 J
- 98.** The perimeter of a rectangle is 54 cm and its length is twice its breadth. Find its area. [\[Perimeter & Area\]](#)
- (A) 243 square cm
 - (B) 180 square cm
 - (C) 162 square cm
 - (D) 324 square cm
 - (E) 81 square cm
 - (F) 54 square cm
 - (G) 108 square cm
- 99.** Rusting goes faster near the sea because salty, moist air speeds it up. A nail rusts fully in 12 days inland but 3 times faster at the coast. In how many days does it rust at the coast? [\[Physical & Chemical Changes\]](#)
- (A) 15 days
 - (B) 36 days
 - (C) 9 days
 - (D) 4 days
 - (E) 6 days
 - (F) 3 days

100. The sum of two rational numbers is $-\frac{3}{4}$. If one of them is $\frac{5}{12}$, find the other. [\[Rational Numbers\]](#)

- (A) $-\frac{1}{3}$
- (B) $\frac{1}{3}$
- (C) $\frac{7}{6}$
- (D) $\frac{14}{12}$
- (E) $-\frac{7}{12}$
- (F) $-\frac{7}{6}$